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Practical 4: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Key Questions / Analysis / Interpretation

1. How are DOM elements selected and modified?
2. Are event listeners attached correctly?
3. Is localStorage used for remembering theme choice?
4. Does interactivity improve usability without breaking accessibility?

Key Skills to be Addressed

DOM selection, event handling, class manipulation, localStorage, basic animation

Applications Interactive dashboards and dynamic frontend behaviour
Learning Outcome Students will apply JavaScript to enhance user experience and client-side

interactivity.

CO Mapping CO3, CO4

Dataset/Test Data Static FAQ/event content

Tools/Technology to be Used

JavaScript ES6+, HTML5, CSS3, browser dev tools

Total Hours Implementation: 4 hours | **Total engagement:** 6 hours

Post Laboratory Work Description

Submit live demo and source-code explanation.

Evaluation Strategy Including Viva

DOM logic, event handling correctness, usability, and browser console testing.

Advanced/Intermediate Extension

Intermediate: Add transition effects to cards/buttons.

Advanced: Store user UI preferences in localStorage and restore them on page load.

Code:

Style.css:

```
/* =====  
Practical 4 – JavaScript DOM & Interactivity  
StudentHub – CSS for all JS components  
===== */  
  
:root {  
  --primary: #1a73e8;  
  --secondary: #0b3d91;  
  --accent: #fbbc04;  
  --dark: #1e2a3a;
```

```
--light: #ffffff;
--bg: #f0f4f9;
--surface: #ffffff;
--muted: #6b7280;
--border: #d1d5db;
--radius: 8px;
--shadow: 0 4px 16px rgba(0,0,0,.12);
--trans: 0.28s ease;
--font: 'Segoe UI', system-ui, sans-serif;
}
```

```
/* ---- Dark Theme overrides ---- */
```

```
body.dark-mode {
```

```
--bg: #0f172a;
```

```
--surface: #1e293b;
```

```
--dark: #f1f5f9;
```

```
--muted: #94a3b8;
```

```
--border: #334155;
```

```
}
```

```
body.dark-mode { background: var(--bg); color: var(--dark); }
```

```
body.dark-mode .navbar { background: #0f172a; }
```

```
body.dark-mode .card-comp { background: var(--surface);
border-color: var(--border); }
```

```
body.dark-mode .faq-item { background: var(--surface);
border-color: var(--border); }
```

```
body.dark-mode .sidebar-comp { background: var(--surface); }
```

```
body.dark-mode .modal-box { background: var(--surface); }
```

```
/* ---- Reset ---- */
```

```
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
```

```
html { scroll-behavior: smooth; }
```

```
body { font-family: var(--font); background: var(--bg); color:
var(--dark); line-height: 1.7; transition: background var(--trans), color
var(--trans); }
```

```
/* =====  
SKIP LINK  
===== */  
.skip-link {  
  position: absolute; top: -50px; left: 0;  
  background: var(--primary); color: #fff;  
  padding: .5rem 1rem; z-index: 9999;  
  text-decoration: none; font-weight: 600; border-radius: 0 0 4px 0;  
  transition: top var(--trans);  
}  
.skip-link:focus { top: 0; }
```

```
/* =====  
NOTIFICATION BANNER  
===== */  
#notification-banner {  
  display: flex;  
  align-items: center;  
  justify-content: space-between;  
  gap: 1rem;  
  background: var(--secondary);  
  color: #fff;  
  padding: .65rem 1.5rem;  
  font-size: .92rem;  
  transition: max-height var(--trans), opacity var(--trans), padding  
var(--trans);  
  overflow: hidden;  
  max-height: 80px;  
}  
#notification-banner.hidden {  
  max-height: 0;  
  opacity: 0;  
  padding-top: 0;  
  padding-bottom: 0;  
}
```

```

#notification-banner a { color: var(--accent); font-weight: 600;
text-decoration: none; }
#notification-banner a:hover { text-decoration: underline; }
#close-banner {
  background: transparent; border: none; color: #fff;
  font-size: 1.3rem; cursor: pointer; line-height: 1;
  padding: 0 .2rem;
  transition: opacity var(--trans);
}
#close-banner:hover { opacity: .7; }

/* =====
  NAVBAR (hamburger controlled by JS)
  ===== */
.navbar {
  background: var(--secondary);
  padding: 0 1.5rem;
  display: flex;
  align-items: center;
  justify-content: space-between;
  height: 62px;
  position: sticky; top: 0; z-index: 500;
  box-shadow: 0 2px 8px rgba(0,0,0,.2);
}
.logo-link { font-size: 1.35rem; font-weight: 700; color: var(--accent);
text-decoration: none; }

/* Hamburger */
.hamburger {
  display: none; flex-direction: column; gap: 5px;
  background: none; border: none; cursor: pointer; padding: 4px;
}
.hamburger span { display: block; width: 25px; height: 2px;
background: #fff; border-radius: 2px; transition: transform
var(--trans), opacity var(--trans); }

```

```
.hamburger[aria-expanded="true"] span:nth-child(1) { transform:
translateY(7px) rotate(45deg); }
.hamburger[aria-expanded="true"] span:nth-child(2) { opacity: 0; }
.hamburger[aria-expanded="true"] span:nth-child(3) { transform:
translateY(-7px) rotate(-45deg); }
```

```
.nav-links { display: flex; gap: 1.4rem; list-style: none; align-items:
center; }
.nav-links a { color: #c8d9f5; text-decoration: none; font-size: .92rem;
border-bottom: 2px solid transparent; padding-bottom: 2px;
transition: color var(--trans), border-color var(--trans); }
.nav-links a:hover, .nav-links a[aria-current="page"] { color:
var(--accent); border-bottom-color: var(--accent); }
```

```
/* Theme toggle button */
#theme-btn {
  background: rgba(255,255,255,.12); border: 1px solid
  rgba(255,255,255,.25);
  color: #fff; border-radius: 999px; padding: .3rem .85rem;
  cursor: pointer; font-size: .85rem; transition: background
  var(--trans);
  white-space: nowrap;
}
#theme-btn:hover { background: rgba(255,255,255,.25); }
```

```
@media (max-width: 768px) {
  .hamburger { display: flex; }
  .nav-links {
    display: none; flex-direction: column;
    position: absolute; top: 62px; left: 0; right: 0;
    background: var(--secondary); padding: 1rem 1.5rem 1.5rem;
    gap: .6rem; box-shadow: 0 4px 12px rgba(0,0,0,.2);
  }
  .nav-links.open { display: flex; }
}
```

```

/* =====
PAGE CONTAINER
===== */
.container { max-width: 1100px; margin: 0 auto; padding: 2rem
1.5rem; }

.page-title {
  font-size: 1.8rem; font-weight: 700; color: var(--secondary);
  margin-bottom: .4rem;
}
body.dark-mode .page-title { color: #93c5fd; }
.page-subtitle { color: var(--muted); margin-bottom: 2rem; }

/* Section headings */
.sec-title {
  font-size: 1.25rem; font-weight: 700;
  color: var(--secondary); margin-bottom: 1rem;
  padding-bottom: .4rem;
  border-bottom: 3px solid var(--accent);
  display: inline-block;
}
body.dark-mode .sec-title { color: #93c5fd; }

/* Grid layout for feature sections */
.features-layout {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 2rem;
}
@media (max-width: 768px) {
  .features-layout { grid-template-columns: 1fr; }
}

.card-comp {

```

```

background: var(--surface);
border-radius: var(--radius);
padding: 1.5rem;
box-shadow: var(--shadow);
border: 1px solid var(--border);
}

/* =====
FAQ ACCORDION (JS controlled)
===== */
.faq-list { list-style: none; }
.faq-item {
background: var(--surface);
border: 1px solid var(--border);
border-radius: var(--radius);
margin-bottom: .7rem;
overflow: hidden;
transition: box-shadow var(--trans);
}
.faq-item:hover { box-shadow: var(--shadow); }

.faq-btn {
width: 100%; text-align: left;
background: none; border: none; cursor: pointer;
padding: 1rem 1.2rem;
font-size: 1rem; font-weight: 600; color: var(--secondary);
display: flex; justify-content: space-between; align-items: center;
gap: .5rem; transition: color var(--trans);
}
body.dark-mode .faq-btn { color: #e2e8f0; }
.faq-btn:hover { color: var(--primary); }

.faq-icon {
font-size: 1.1rem; transition: transform var(--trans);
flex-shrink: 0; color: var(--primary);

```

```
}  
.faq-btn[aria-expanded="true"] .faq-icon { transform: rotate(45deg); }
```

```
.faq-answer {  
  max-height: 0; overflow: hidden;  
  transition: max-height .35s ease, padding var(--trans);  
  padding: 0 1.2rem;  
  color: var(--muted); font-size: .95rem;  
}
```

```
.faq-answer.open {  
  max-height: 400px;  
  padding: .2rem 1.2rem 1rem;  
}
```

```
/* =====  
  MODAL POPUP  
  ===== */
```

```
.modal-overlay {  
  position: fixed; inset: 0;  
  background: rgba(0,0,0,.55);  
  display: flex; align-items: center; justify-content: center;  
  z-index: 1000; opacity: 0; pointer-events: none;  
  transition: opacity var(--trans);  
  padding: 1rem;  
}
```

```
.modal-overlay.open { opacity: 1; pointer-events: all; }
```

```
.modal-box {  
  background: var(--surface); border-radius: 12px;  
  padding: 2rem; max-width: 500px; width: 100%;  
  box-shadow: 0 20px 60px rgba(0,0,0,.3);  
  transform: translateY(20px) scale(.97);  
  transition: transform var(--trans);  
  position: relative;  
}
```

```
.modal-overlay.open .modal-box { transform: translateY(0) scale(1); }
```

```
.modal-close {  
  position: absolute; top: .8rem; right: 1rem;  
  background: none; border: none; font-size: 1.5rem;  
  cursor: pointer; color: var(--muted);  
  transition: color var(--trans);  
}  
.modal-close:hover { color: var(--dark); }
```

```
.modal-box h2 { margin-bottom: 1rem; color: var(--secondary); }  
body.dark-mode .modal-box h2 { color: #93c5fd; }  
.modal-box p { color: var(--muted); margin-bottom: 1.2rem; }  
.modal-box .btn { display: inline-block; padding: .6rem 1.4rem;  
background: var(--primary); color: #fff; border-radius: var(--radius);  
border: none; cursor: pointer; font-size: .95rem; transition:  
background var(--trans); text-decoration: none; }  
.modal-box .btn:hover { background: #1558b0; }
```

```
/* =====  
  CONTENT SLIDER  
  ===== */  
.slider-wrap {  
  position: relative; overflow: hidden;  
  border-radius: var(--radius);  
  background: var(--surface);  
  border: 1px solid var(--border);  
}  
.slider-track {  
  display: flex;  
  transition: transform .45s cubic-bezier(.4,0,.2,1);  
}  
.slide {  
  min-width: 100%;  
  padding: 2rem 1.5rem;
```

```

    text-align: center;
}
.slide .slide-icon { font-size: 3rem; margin-bottom: .8rem; }
.slide h3 { color: var(--secondary); margin-bottom: .5rem; }
body.dark-mode .slide h3 { color: #93c5fd; }
.slide p { color: var(--muted); font-size: .95rem; }

.slider-controls {
    display: flex; align-items: center; justify-content: center;
    gap: 1rem; padding: 1rem;
    border-top: 1px solid var(--border);
}
.slider-btn {
    background: var(--primary); color: #fff;
    border: none; border-radius: 50%;
    width: 36px; height: 36px; font-size: 1rem;
    cursor: pointer; transition: background var(--trans);
    display: flex; align-items: center; justify-content: center;
}
.slider-btn:hover { background: var(--secondary); }
.slider-dots { display: flex; gap: .5rem; }
.dot {
    width: 10px; height: 10px; border-radius: 50%;
    background: var(--border); border: none; cursor: pointer;
    transition: background var(--trans);
}
.dot.active { background: var(--primary); }

/* =====
   TRANSITION CARDS (Intermediate Extension)
   ===== */
.trans-cards { display: grid; grid-template-columns: repeat(auto-fit,
minmax(180px, 1fr)); gap: 1rem; margin-top: 1rem; }
.trans-card {
    background: var(--surface);

```

```

border: 1px solid var(--border);
border-radius: var(--radius);
padding: 1.2rem;
text-align: center;
cursor: pointer;
transition: transform .25s ease, box-shadow .25s ease, border-color
.25s ease;
}
.trans-card:hover, .trans-card:focus {
  transform: translateY(-6px);
  box-shadow: 0 8px 24px rgba(26,115,232,.2);
  border-color: var(--primary);
  outline: none;
}
.trans-card .tc-icon { font-size: 2rem; margin-bottom: .5rem; }
.trans-card h3 { font-size: .95rem; color: var(--secondary); }
body.dark-mode .trans-card h3 { color: #e2e8f0; }

```

```

/* =====
FOOTER
===== */

```

```

.site-footer {
  background: var(--dark);
  color: #94a3b8;
  text-align: center;
  padding: 1.5rem;
  margin-top: 3rem;
  font-size: .88rem;
}
body.dark-mode .site-footer { background: #0f172a; }
.site-footer a { color: var(--accent); text-decoration: none; }

```

```

/* =====
UTILITY
===== */

```

```

.visually-hidden { position: absolute; width: 1px; height: 1px; padding:
0; margin: -1px; overflow: hidden; clip: rect(0,0,0,0); white-space:
nowrap; border: 0; }
.mt-1 { margin-top: 1rem; }
.mt-2 { margin-top: 2rem; }
.open-modal-btn {
  display: inline-block; padding: .6rem 1.4rem;
  background: var(--primary); color: #fff; border: none;
  border-radius: var(--radius); cursor: pointer; font-size: .95rem;
  transition: background var(--trans), transform var(--trans);
}
.open-modal-btn:hover { background: var(--secondary); transform:
translateY(-2px); }

```

main.js

```

/**
 * Practical 4 – JavaScript DOM Manipulation,
 * Event Handling & UI Interactivity
 * StudentHub
 *
 * Features implemented:
 * 1. Notification banner (dismiss + auto-hide)
 * 2. Hamburger / mobile nav toggle
 * 3. Light / Dark theme switcher (localStorage)
 * 4. FAQ accordion (collapsible, keyboard-friendly)
 * 5. Modal popup (open / close / Escape key / focus trap)
 * 6. Content / image slider (prev/next/dots/auto-play)
 * 7. Transition effects on cards (hover via CSS, click via JS)
 * 8. localStorage – restore theme preference on page load
 */

'use strict';

/* =====
1. THEME SWITCHER (localStorage)

```

```

===== */
const THEME_KEY = 'sh_theme';
const themeBtn = document.getElementById('theme-btn');
const body = document.body;

/**
 * Apply the given theme ('dark' | 'light') and persist it.
 * @param {string} theme
 */
function applyTheme(theme) {
  if (theme === 'dark') {
    body.classList.add('dark-mode');
    if (themeBtn) themeBtn.textContent = '☀️ Light Mode';
    if (themeBtn) themeBtn.setAttribute('aria-label', 'Switch to light mode');
  } else {
    body.classList.remove('dark-mode');
    if (themeBtn) themeBtn.textContent = '🌙 Dark Mode';
    if (themeBtn) themeBtn.setAttribute('aria-label', 'Switch to dark mode');
  }
  localStorage.setItem(THEME_KEY, theme);
}

// Restore saved preference on load
(function restoreTheme() {
  const saved = localStorage.getItem(THEME_KEY) || 'light';
  applyTheme(saved);
})();

// Toggle on button click
if (themeBtn) {
  themeBtn.addEventListener('click', () => {
    const current = localStorage.getItem(THEME_KEY) || 'light';
    applyTheme(current === 'dark' ? 'light' : 'dark');
  });
}

/* =====
2. NOTIFICATION BANNER
===== */
const banner = document.getElementById('notification-banner');

```

```

const closeBtn = document.getElementById('close-banner');
const BANNER_KEY = 'sh_banner_dismissed';

// If already dismissed in this session, hide immediately
if (banner && sessionStorage.getItem(BANNER_KEY)) {
  banner.classList.add('hidden');
}

if (closeBtn) {
  closeBtn.addEventListener('click', () => {
    banner.classList.add('hidden');
    sessionStorage.setItem(BANNER_KEY, '1');
  });
}

/* =====
3. HAMBURGER NAV
===== */
const hamburger = document.getElementById('hamburger-btn');
const navLinks = document.getElementById('main-nav');

if (hamburger && navLinks) {
  hamburger.addEventListener('click', () => {
    const expanded = hamburger.getAttribute('aria-expanded') === 'true';
    hamburger.setAttribute('aria-expanded', String(!expanded));
    navLinks.classList.toggle('open');
  });

  // Close nav on outside click
  document.addEventListener('click', (e) => {
    if (!hamburger.contains(e.target) && !navLinks.contains(e.target)) {
      hamburger.setAttribute('aria-expanded', 'false');
      navLinks.classList.remove('open');
    }
  });
}

/* =====
4. FAQ ACCORDION
===== */

```

```

const faqBtns = document.querySelectorAll('.faq-btn');

faqBtns.forEach((btn) => {
  btn.addEventListener('click', () => {
    const expanded = btn.getAttribute('aria-expanded') === 'true';
    const answer = document.getElementById(btn.getAttribute('aria-controls'));

    // Close all others (single-open accordion)
    faqBtns.forEach((b) => {
      if (b !== btn) {
        b.setAttribute('aria-expanded', 'false');
        const a = document.getElementById(b.getAttribute('aria-controls'));
        if (a) a.classList.remove('open');
      }
    });

    // Toggle current
    btn.setAttribute('aria-expanded', String(!expanded));
    if (answer) answer.classList.toggle('open', !expanded);
  });

  // Keyboard: Space / Enter already fire click on <button>
  // Arrow Up/Down to navigate between FAQ items
  btn.addEventListener('keydown', (e) => {
    const items = [...faqBtns];
    const idx = items.indexOf(btn);
    if (e.key === 'ArrowDown') { e.preventDefault(); items[Math.min(idx + 1,
items.length - 1)].focus(); }
    if (e.key === 'ArrowUp') { e.preventDefault(); items[Math.max(idx - 1,
0)].focus(); }
  });
});

/* =====
5. MODAL POPUP
===== */
const modalOverlay = document.getElementById('modal-overlay');
const modalCloseBtn = document.getElementById('modal-close');
const openModalBtns = document.querySelectorAll('.open-modal-btn');

```

```

// All focusable elements inside modal
function getFocusable(container) {
  return [...container.querySelectorAll(
    'a[href], button:not([disabled]), input, textarea, select,
    [tabindex]:not([tabindex="-1"])'
  )];
}

function openModal() {
  if (!modalOverlay) return;
  modalOverlay.classList.add('open');
  modalOverlay.removeAttribute('hidden');
  // Move focus to first focusable inside modal
  const focusable = getFocusable(modalOverlay);
  if (focusable.length) focusable[0].focus();
}

function closeModal() {
  if (!modalOverlay) return;
  modalOverlay.classList.remove('open');
  modalOverlay.setAttribute('hidden', "");
  // Return focus to the button that opened the modal
  if (lastFocused) lastFocused.focus();
}

let lastFocused = null;

openModalBtns.forEach((btn) => {
  btn.addEventListener('click', () => {
    lastFocused = btn;
    openModal();
  });
});

if (modalCloseBtn) modalCloseBtn.addEventListener('click', closeModal);

// Close on overlay background click
if (modalOverlay) {
  modalOverlay.addEventListener('click', (e) => {
    if (e.target === modalOverlay) closeModal();
  });
}

```

```
});  
}
```

```
// Close on Escape; trap Tab inside modal  
document.addEventListener('keydown', (e) => {  
  if (!modalOverlay || !modalOverlay.classList.contains('open')) return;  
  
  if (e.key === 'Escape') {  
    closeModal();  
    return;  
  }  
});
```

```
// Focus trap  
if (e.key === 'Tab') {  
  const focusable = getFocusable(modalOverlay);  
  if (!focusable.length) return;  
  const first = focusable[0];  
  const last = focusable[focusable.length - 1];  
  
  if (e.shiftKey && document.activeElement === first) {  
    e.preventDefault(); last.focus();  
  } else if (!e.shiftKey && document.activeElement === last) {  
    e.preventDefault(); first.focus();  
  }  
}  
});
```

```
/* =====  
6. CONTENT SLIDER  
===== */
```

```
const sliderTrack = document.getElementById('slider-track');  
const prevBtn = document.getElementById('slider-prev');  
const nextBtn = document.getElementById('slider-next');  
const dotsWrap = document.getElementById('slider-dots');
```

```
let currentSlide = 0;  
let autoPlayTimer = null;  
const AUTOPLAY_DELAY = 4000;
```

```
function getSlides() {
```

```

    return sliderTrack ? [...sliderTrack.querySelectorAll('.slide')] : [];
}

function getDots() {
    return dotsWrap ? [...dotsWrap.querySelectorAll('.dot')] : [];
}

function goToSlide(index) {
    const slides = getSlides();
    if (!slides.length || !sliderTrack) return;

    currentSlide = (index + slides.length) % slides.length;
    sliderTrack.style.transform = `translateX(-${currentSlide * 100}%)`;

    // Update dots
    getDots().forEach((d, i) => {
        d.classList.toggle('active', i === currentSlide);
        d.setAttribute('aria-selected', String(i === currentSlide));
    });

    // Update live region for screen readers
    const liveEl = document.getElementById('slider-live');
    if (liveEl) liveEl.textContent = `Slide ${currentSlide + 1} of ${slides.length}`;
}

function startAutoPlay() {
    stopAutoPlay();
    autoPlayTimer = setInterval(() => goToSlide(currentSlide + 1),
    AUTOPLAY_DELAY);
}

function stopAutoPlay() {
    if (autoPlayTimer) { clearInterval(autoPlayTimer); autoPlayTimer = null; }
}

if (prevBtn) prevBtn.addEventListener('click', () => { stopAutoPlay();
goToSlide(currentSlide - 1); startAutoPlay(); });
if (nextBtn) nextBtn.addEventListener('click', () => { stopAutoPlay();
goToSlide(currentSlide + 1); startAutoPlay(); });

```

```

// Build dots dynamically
(function initSlider() {
  const slides = getSlides();
  if (!slides.length || !dotsWrap) return;

  slides.forEach((_, i) => {
    const dot = document.createElement('button');
    dot.className = 'dot' + (i === 0 ? ' active' : '');
    dot.setAttribute('aria-label', `Go to slide ${i + 1}`);
    dot.setAttribute('aria-selected', String(i === 0));
    dot.addEventListener('click', () => { stopAutoPlay(); goToSlide(i);
startAutoPlay(); });
    dotsWrap.appendChild(dot);
  });

  goToSlide(0);
  startAutoPlay();

  // Pause autoplay on hover / focus
  const wrap = document.querySelector('.slider-wrap');
  if (wrap) {
    wrap.addEventListener('mouseenter', stopAutoPlay);
    wrap.addEventListener('focusin', stopAutoPlay);
    wrap.addEventListener('mouseleave', startAutoPlay);
    wrap.addEventListener('focusout', startAutoPlay);
  }
})();

// Keyboard navigation on slider
if (sliderTrack) {
  sliderTrack.setAttribute('tabindex', '0');
  sliderTrack.addEventListener('keydown', (e) => {
    if (e.key === 'ArrowLeft') { stopAutoPlay(); goToSlide(currentSlide - 1);
startAutoPlay(); }
    if (e.key === 'ArrowRight') { stopAutoPlay(); goToSlide(currentSlide + 1);
startAutoPlay(); }
  });
}

/* =====

```

7. TRANSITION CARD CLICK EFFECT (Intermediate Ext.)

```
===== */
document.querySelectorAll('.trans-card').forEach((card) => {
  card.addEventListener('click', () => {
    // Pulse animation via class
    card.classList.add('clicked');
    card.addEventListener('animationend', () => card.classList.remove('clicked'), {
      once: true });
    // Visual feedback: briefly highlight border
    card.style.borderColor = 'var(--accent)';
    setTimeout(() => { card.style.borderColor = ''; }, 600);
  });

  // Accessible keyboard activation
  card.setAttribute('tabindex', '0');
  card.addEventListener('keydown', (e) => {
    if (e.key === 'Enter' || e.key === ' ') { e.preventDefault(); card.click(); }
  });
});
```

```
/* =====
8. RESTORE UI PREFERENCES ON PAGE LOAD (Advanced Ext.)
   Theme already restored at top. Add any other prefs here.
===== */
```

```
(function restoreUIPrefs() {
  // Example: restore font-size preference
  const fontSize = localStorage.getItem('sh_font_size');
  if (fontSize) document.documentElement.style.fontSize = fontSize;
})();
```

```
/* =====
HELPER: Show a toast-style notification
===== */
```

```
/**
```

```
 * Display a temporary toast message.
 * @param {string} message
 * @param {'success'|'error'|'info'} type
 */
```

```
function showToast(message, type = 'info') {
  let toast = document.getElementById('toast-container');
```

```

if (!toast) {
  toast = document.createElement('div');
  toast.id = 'toast-container';
  toast.setAttribute('aria-live', 'polite');
  toast.setAttribute('role', 'status');
  Object.assign(toast.style, {
    position: 'fixed', bottom: '1.5rem', right: '1.5rem',
    zIndex: '2000', display: 'flex', flexDirection: 'column', gap: '.5rem'
  });
  document.body.appendChild(toast);
}

```

```

const colors = { success: '#16a34a', error: '#dc2626', info: '#1a73e8' };
const msg = document.createElement('div');
msg.textContent = message;
Object.assign(msg.style, {
  background: colors[type] || colors.info,
  color: '#fff', padding: '.7rem 1.2rem',
  borderRadius: '8px', fontSize: '.9rem',
  boxShadow: '0 4px 12px rgba(0,0,0,.2)',
  opacity: '0', transition: 'opacity .3s ease',
  maxWidth: '320px'
});
toast.appendChild(msg);
requestAnimationFrame(() => { msg.style.opacity = '1'; });
setTimeout(() => {
  msg.style.opacity = '0';
  msg.addEventListener('transitionend', () => msg.remove(), { once: true });
}, 3500);
}

```

```

// Expose showToast globally for inline use
window.showToast = showToast;

```

Index.html

```

<!DOCTYPE html>
<html lang="en">

```

```

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Practical 4 – JavaScript DOM
Manipulation, Event Handling, and UI Interactivity for StudentHub." />
  <title>Practical 4 – JS Interactivity | StudentHub</title>
  <link rel="stylesheet" href="css/style.css" />
  <style>
    /* Pulse keyframe for card click feedback */
    @keyframes pulse {
      0% { transform: scale(1); }
      50% { transform: scale(1.04); }
      100% { transform: scale(1); }
    }
    .trans-card.clicked { animation: pulse .35s ease; }
  </style>
</head>
<body>

  <!-- Skip to content (Accessibility) -->
  <a class="skip-link" href="#main-content">Skip to main content</a>

  <!-- =====
  FEATURE 1: NOTIFICATION BANNER
  ===== -->
  <div id="notification-banner" role="alert" aria-live="polite">
    <span>🎓 <strong>Semester IV Registration</strong> is now open!
    Deadline: 25 Aug 2026.
    <a href="#">Register Now →</a>
  </span>
  <button id="close-banner" aria-label="Dismiss notification">✕</button>
</div>

  <!-- =====
  FEATURE 2: NAVBAR with HAMBURGER MENU
  ===== -->
  <header>

```

```
<nav class="navbar" aria-label="Primary navigation">
  <a href="index.html" class="logo-link" aria-label="StudentHub
Home">🎓 StudentHub</a>
```

```
<!-- FEATURE 3: THEME SWITCHER BUTTON -->
<button id="theme-btn" aria-label="Switch to dark mode">🌙 Dark
Mode</button>
```

```
<button class="hamburger" id="hamburger-btn"
  aria-controls="main-nav"
  aria-expanded="false"
  aria-label="Toggle navigation menu">
  <span></span><span></span><span></span>
</button>
```

```
<ul class="nav-links" id="main-nav" role="list">
  <li><a href="index.html" aria-current="page">Home</a></li>
  <li><a href="#faq-section">FAQ</a></li>
  <li><a href="#slider-section">Announcements</a></li>
  <li><a href="#modal-section">Events</a></li>
  <li><a href="#cards-section">Courses</a></li>
  <li><a href=" ../practical2/login.html">Login</a></li>
</ul>
</nav>
</header>
```

```
<!-- =====
MAIN CONTENT
===== -->
<main id="main-content">
  <div class="container">
```

```
    <h1 class="page-title">Practical 4 – JS DOM Interactivity</h1>
    <p class="page-subtitle">
      Demonstrates: Notification Banner · Hamburger Nav · Dark/Light
Theme ·
```

FAQ Accordion · Modal Popup · Content Slider · Transition Cards ·
localStorage

```
<!-- =====  
      FEATURE 4 + 5: FAQ ACCORDION + MODAL DEMO  
      ===== -->  
<div class="features-layout">
```

```
  <!-- ---- FAQ ACCORDION ---- -->  
  <section id="faq-section" aria-labelledby="faq-heading">  
    <h2 id="faq-heading" class="sec-title">FAQ – Collapsible  
Accordion</h2>  
    <p style="font-size:.88rem;color:var(--muted);margin-bottom:1rem;">  
      Click a question to expand. Use ↑ ↓ arrow keys to navigate.  
    </p>
```

```
    <ul class="faq-list" role="list">  
  
      <li class="faq-item">  
        <button class="faq-btn" id="faq-btn-1"  
          aria-expanded="false"  
          aria-controls="faq-ans-1">  
          How do I register on StudentHub?  
          <span class="faq-icon" aria-hidden="true">+</span>  
        </button>  
        <div class="faq-answer" id="faq-ans-1" role="region"  
aria-labelledby="faq-btn-1">
```

Click Register in the nav. Fill your details, set a
password, and submit. You'll receive a confirmation email after admin
approval.

```
      </div>  
    </li>
```

```
    <li class="faq-item">  
      <button class="faq-btn" id="faq-btn-2"  
        aria-expanded="false"
```

```
        aria-controls="faq-ans-2">
        I forgot my password. What should I do?
        <span class="faq-icon" aria-hidden="true">+</span>
    </button>
    <div class="faq-answer" id="faq-ans-2" role="region"
aria-labelledby="faq-btn-2">
        Go to the Login page and click <em>Forgot your
password?</em>. Enter your registered email and a reset link will be sent
within 5 minutes.
    </div>
</li>
```

```
<li class="faq-item">
    <button class="faq-btn" id="faq-btn-3"
        aria-expanded="false"
        aria-controls="faq-ans-3">
        When are semester results published?
        <span class="faq-icon" aria-hidden="true">+</span>
    </button>
    <div class="faq-answer" id="faq-ans-3" role="region"
aria-labelledby="faq-btn-3">
        Results are published within 30 days after exams. You'll receive
an email notification when they're available on your Dashboard.
    </div>
</li>
```

```
<li class="faq-item">
    <button class="faq-btn" id="faq-btn-4"
        aria-expanded="false"
        aria-controls="faq-ans-4">
        How do I update my personal information?
        <span class="faq-icon" aria-hidden="true">+</span>
    </button>
    <div class="faq-answer" id="faq-ans-4" role="region"
aria-labelledby="faq-btn-4">
```

Log in and go to **My Profile**. You can update mobile number and address directly. For name/DOB changes, contact the admin office.

</div>

<li class="faq-item">

<button class="faq-btn" id="faq-btn-5"

aria-expanded="false"

aria-controls="faq-ans-5">

Is my personal data secure on this portal?

+

</button>

<div class="faq-answer" id="faq-ans-5" role="region"

aria-labelledby="faq-btn-5">

Yes. StudentHub uses HTTPS and stores passwords in hashed form. We do not share your data with third parties. See our Privacy Policy for details.

</div>

</section>

<!-- ---- MODAL POPUP ---- -->

<section id="modal-section" aria-labelledby="modal-sec-heading">

<h2 id="modal-sec-heading" class="sec-title">Modal Popup</h2>

<p style="font-size:.88rem;color:var(--muted);margin-bottom:1rem;">

Opens a dialog with focus trap and Escape key support.

</p>

<div class="card-comp">

<h3 style="margin-bottom:.8rem;">TechSpark Hackathon 2026</h3>

<p

style="color:var(--muted);font-size:.93rem;margin-bottom:1rem;">

A 24-hour inter-college hackathon — AI, IoT, FinTech. Teams of 2–4.

Prizes worth ₹1,00,000.

```
</p>
<button class="open-modal-btn" aria-haspopup="dialog">
  View Details & Register
</button>
</div>
```

```
<div class="card-comp mt-1">
  <h3 style="margin-bottom:.8rem;">Web Dev Workshop</h3>
  <p
style="color:var(--muted);font-size:.93rem;margin-bottom:1rem;">
    Hands-on session covering HTML5, CSS3 & JavaScript. Open
to all years. Free entry.
  </p>
  <button class="open-modal-btn" aria-haspopup="dialog">
    View Details
  </button>
</div>
</section>
```

```
</div><!-- /features-layout -->
```

```
<!-- =====
FEATURE 6: CONTENT SLIDER
===== -->
<section id="slider-section" class="mt-2"
aria-labelledby="slider-heading">
  <h2 id="slider-heading" class="sec-title">Announcements Slider</h2>
  <p style="font-size:.88rem;color:var(--muted);margin-bottom:1rem;">
    Auto-advances every 4 s. Use ← → arrows or dots. Pauses on
hover/focus.
  </p>

  <!-- Screen reader live region -->
  <div id="slider-live" class="visually-hidden" aria-live="polite"
aria-atomic="true"></div>
```

```
<div class="slider-wrap" aria-roledescription="carousel"
aria-label="Campus announcements">
  <div class="slider-track" id="slider-track" aria-live="off">
```

```
    <div class="slide" role="group" aria-roledescription="slide"
aria-label="Slide 1 of 5">
```

```
      <div class="slide-icon" aria-hidden="true"> </div>
```

```
      <h3>Semester IV Registration Open</h3>
```

```
      <p>Log in to select your subjects. Deadline: <strong>25 August
2026</strong>. Late registrations not accepted.</p>
    </div>
```

```
    <div class="slide" role="group" aria-roledescription="slide"
aria-label="Slide 2 of 5">
```

```
      <div class="slide-icon" aria-hidden="true"> </div>
```

```
      <h3>TechSpark Hackathon 2026</h3>
```

```
      <p>Register before <strong>5 September</strong>. 24-hour event.
Prizes worth ₹1,00,000. Form teams of 2–4.</p>
    </div>
```

```
    <div class="slide" role="group" aria-roledescription="slide"
aria-label="Slide 3 of 5">
```

```
      <div class="slide-icon" aria-hidden="true"> </div>
```

```
      <h3>Annual Cultural Fest – Rang 2026</h3>
```

```
      <p>Music, dance, drama & art competitions. <strong>5–7
October 2026.</strong> 30+ colleges participating.</p>
    </div>
```

```
    <div class="slide" role="group" aria-roledescription="slide"
aria-label="Slide 4 of 5">
```

```
      <div class="slide-icon" aria-hidden="true"> </div>
```

```
      <h3>Career Fair 2026</h3>
```

```
      <p>Meet 50+ top recruiters. <strong>15 November 2026.</strong>
Bring your resume. Limited seats — register early!</p>
    </div>
```

```

    <div class="slide" role="group" aria-roledescription="slide"
aria-label="Slide 5 of 5">
      <div class="slide-icon" aria-hidden="true">🏆</div>
      <h3>Sports Day 2026</h3>
      <p>Cricket, football, track & field, badminton. <strong>1
December 2026.</strong> Open to all students.</p>
    </div>

```

```

</div>

```

```

    <div class="slider-controls">
      <button class="slider-btn" id="slider-prev" aria-label="Previous
slide">⏮️</button>
      <div class="slider-dots" id="slider-dots" role="tablist"
aria-label="Select slide"></div>
      <button class="slider-btn" id="slider-next" aria-label="Next
slide">⏭️</button>
    </div>
  </div>
</section>

```

```

<!-- =====
FEATURE 7: TRANSITION CARDS (Intermediate)
===== -->
<section id="cards-section" class="mt-2"
aria-labelledby="cards-heading">
  <h2 id="cards-heading" class="sec-title">Course Cards – Hover
& Click Transitions</h2>
  <p style="font-size:.88rem;color:var(--muted);margin-bottom:.5rem;">
    Hover for lift effect. Click for pulse animation. Keyboard-accessible
via Tab + Enter/Space.
  </p>

  <div class="trans-cards">
    <div class="trans-card" role="button" tabindex="0"
      aria-label="Web Technologies course"
      onclick="showToast('Web Technologies selected!','success')">

```

```

<div class="tc-icon" aria-hidden="true"><img alt="Globe icon" data-bbox="598 93 625 113"/></div>
<h3>Web Technologies</h3>
<p style="font-size:.82rem;color:var(--muted);">HTML5 · CSS3 ·
JavaScript</p>
</div>
<div class="trans-card" role="button" tabindex="0"
  aria-label="Database Management course"
  onclick="showToast('Database Management
selected!','success')">
  <div class="tc-icon" aria-hidden="true"><img alt="Database icon" data-bbox="598 288 625 308"/></div>
  <h3>Database Management</h3>
  <p style="font-size:.82rem;color:var(--muted);">SQL · ER Diagrams ·
Normalization</p>
</div>
<div class="trans-card" role="button" tabindex="0"
  aria-label="Data Structures course"
  onclick="showToast('Data Structures selected!','success')">
  <div class="tc-icon" aria-hidden="true"><img alt="Tree icon" data-bbox="598 463 625 483"/></div>
  <h3>Data Structures</h3>
  <p style="font-size:.82rem;color:var(--muted);">Arrays · Trees ·
Graphs</p>
</div>
<div class="trans-card" role="button" tabindex="0"
  aria-label="Operating Systems course"
  onclick="showToast('Operating Systems selected!','success')">
  <div class="tc-icon" aria-hidden="true"><img alt="Laptop icon" data-bbox="598 638 625 658"/></div>
  <h3>Operating Systems</h3>
  <p style="font-size:.82rem;color:var(--muted);">Process · Memory ·
File Systems</p>
</div>
<div class="trans-card" role="button" tabindex="0"
  aria-label="Python Programming course"
  onclick="showToast('Python Programming selected!','success')">
  <div class="tc-icon" aria-hidden="true"><img alt="Snake icon" data-bbox="598 808 625 828"/></div>
  <h3>Python Programming</h3>
  <p style="font-size:.82rem;color:var(--muted);">OOP · Libraries ·
Projects</p>

```

```

</div>
<div class="trans-card" role="button" tabindex="0"
  aria-label="Mathematics course"
  onclick="showToast('Mathematics selected!','success')">
  <div class="tc-icon" aria-hidden="true">📐 </div>
  <h3>Mathematics – II</h3>
  <p style="font-size:.82rem;color:var(--muted);">Calculus · Linear
Algebra</p>
</div>
</div>
</section>

</div><!-- /container -->
</main>

<!-- =====
MODAL DIALOG (hidden by default)
===== -->
<div id="modal-overlay" class="modal-overlay" role="dialog"
  aria-modal="true" aria-labelledby="modal-title" hidden>
  <div class="modal-box">
    <button class="modal-close" id="modal-close" aria-label="Close
dialog">✕ </button>
    <h2 id="modal-title">Event Registration</h2>
    <p>
      <strong>TechSpark Hackathon 2026</strong><br />
      📅 10–11 September 2026 &nbsp;  📍 College Auditorium,
Pune<br />
      Teams of 2–4 · Categories: AI, IoT, FinTech<br />
      Prize Pool: <strong>₹1,00,000</strong>
    </p>
    <p>Fill in your details below to register your team.</p>
    <form action="#" method="post" aria-label="Event registration form"
novalidate>
      <div style="margin-bottom:.8rem;">

```

```

        <label for="m-name"
style="display:block;font-weight:600;margin-bottom:.3rem;">Team
Name</label>
        <input type="text" id="m-name" name="teamName"
        style="width:100%;padding:.5rem .8rem;border:1.5px solid
var(--border);border-radius:6px;font-size:.95rem;" />
    </div>
    <div style="margin-bottom:1rem;">
        <label for="m-email"
style="display:block;font-weight:600;margin-bottom:.3rem;">Contact
Email</label>
        <input type="email" id="m-email" name="email"
        style="width:100%;padding:.5rem .8rem;border:1.5px solid
var(--border);border-radius:6px;font-size:.95rem;" />
    </div>
    <button type="submit" class="btn"
        onclick="closeModal(); showToast('Registration
submitted!','success'); return false;">
        Register Team
    </button>
    <button type="button" class="btn"
        style="background:var(--muted);margin-left:.5rem;"
        onclick="closeModal()">Cancel</button>
</form>
</div>
</div>

<!-- =====
FOOTER
===== -->
<footer class="site-footer" role="contentinfo">
    <p>&copy; 2026 StudentHub – Practical 4: JS DOM Manipulation &
Interactivity</p>
    <p><a href="../practical2/index.html">← Practical 2</a> &nbsp;|&nbsp;
    <a href="../practical3/index.html">← Practical 3</a> &nbsp;|&nbsp;
    <a href="../practical5/index.html">Practical 5 →</a></p>
</footer>

```

```
<!-- Main JS (deferred to run after DOM is ready) -->
```

```
<script src="js/main.js"></script>
```

```
<!-- Inline: expose closeModal to HTML onclick attribute -->
```

```
<script>
```

```
  // Make closeModal accessible from inline onclick handlers in the modal  
  function closeModal() {
```

```
    const overlay = document.getElementById('modal-overlay');
```

```
    if (overlay) {
```

```
      overlay.classList.remove('open');
```

```
      overlay.setAttribute('hidden', '');
```

```
      // Return focus to last focused element tracked by main.js
```

```
      if (window._lastFocused) window._lastFocused.focus();
```

```
    }
```

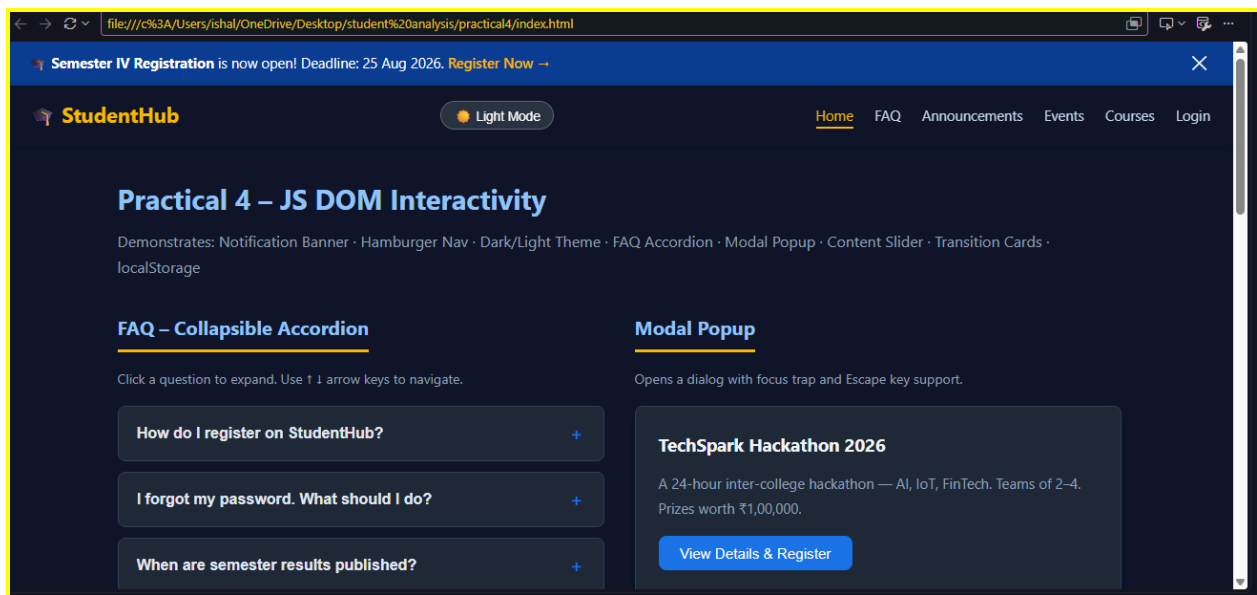
```
  }
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



FAQ – Collapsible Accordion

Click a question to expand. Use ↑ ↓ arrow keys to navigate.

How do I register on StudentHub?



I forgot my password. What should I do?



When are semester results published?



How do I update my personal information?



Is my personal data secure on this portal?



Modal Popup

Opens a dialog with focus trap and Escape key support.

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[View Details & Register](#)

Web Dev Workshop

Hands-on session covering HTML5, CSS3 & JavaScript. Open to all years. Free entry.

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Announcements Slider

Auto-advances every 4 s. Use ← → arrows or dots. Pauses on hover/focus.



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Web Technologies

HTML5 · CSS3 · JavaScript



Database Management

SQL · ER Diagrams · Normalization



Data Structures

Arrays · Trees · Graphs



Operating Systems

Process · Memory · File Systems



Python Programming

OOP · Libraries · Projects



Mathematics – II

Calculus · Linear Algebra

Sign In

Don't have an account? [Register here](#).

Email Address *

Password *

☐

Remember me on this device

[Forgot your password?](#)
